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Biochemistry Module, 1st Year Medicine
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Fatty Acid Metabolism

Course Objectives

- List the enzymes involved in the digestion of dietary lipids.

- Determine the characteristics of malonyl-CoA biosynthesis and its regulation.

- Describe the three characteristics (cellular compartment, enzyme, and energy balance) of the fatty acid activation step to acyl-CoA.

- Describe the mechanism and enzymes used to ensure the transfer of acyl-CoA from the cytosol to the mitochondria via carnitine.

- List the four enzymatic steps, with their coenzymes, of one turn of the Lynen helix.

- Describe the chemical and energy balance of one turn of the Lynen helix.

- Evaluate the chemical and energy balance of the β-oxidation of fatty acids, according to their number of carbon atoms and double bonds.

- Describe the three origins and four metabolic pathways used by acetyl-CoA.
- List all the enzymes and coenzymes involved in every step of fatty acid biosynthesis.

- Identify the cellular compartments involved in the biosynthesis of short-, medium-, and long-chain fatty acids.

- Describe the metabolic, enzymatic, and hormonal regulation of fatty acid biosynthesis.

I/Digestion of dietary lipids and mobilization of stored lipids:

A/Digestion of dietary lipids:

1) The main lipids in the human and animal diet consist mainly of triglycerides, phospholipids and sterols.

- The digestion of these lipids depends on pancreatic enzymes and bile salts.
- The enzymes that hydrolyze lipids are lipases and phospholipases. Their activity takes place in the small intestine.

- The complete action of **triglyceride lipase** (pancreatic) leads to the release of **2 fatty acids and 2-monoacylglycerol.**

- There are **four phospholipases** that hydrolyze phospholipids: A1, A2, C and D.

- The action of **bile salts** completes the emulsification process and the formation of micelles

2) **Mixed micelles** contain fatty acids and 2-monoacylglycerols; they are absorbed by enterocytes (absorptive cells of the small intestine).

- Once inside the enterocyte, the fatty acids are taken up by a specific transporter that transports them to the **smooth endoplasmic reticulum**. They are joined by the 2-monoacylglycerols, which are able to cross the membrane by passive diffusion.
- The fatty acids and 2-monoacylglycerols are recombined into triacylglycerols by enzymes in the smooth endoplasmic reticulum.

3) **Lipoproteins** are the transport forms of hydrophobic fats in blood plasma.

The main lipoproteins are:

- Chylomicrons, synthesized in enterocytes (the intestine): the transport form of dietary triglycerides to user tissues and adipose tissue.

- **VLDL** (very low-density lipoproteins) synthesised in the liver

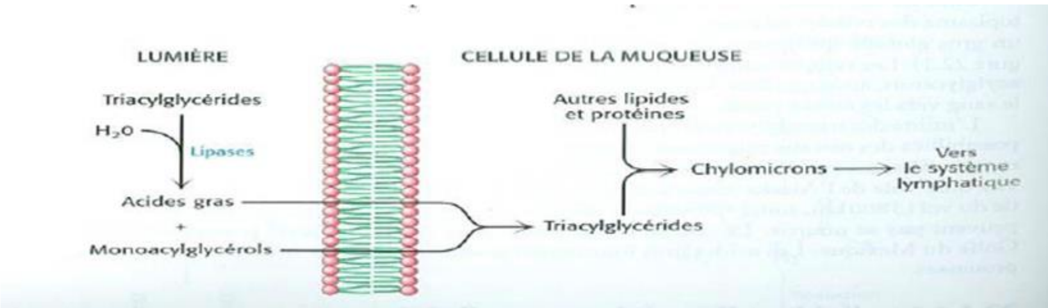
- **LDL** (low-density lipoproteins)

- **HDL** (high-density lipoproteins) synthesised in the blood

- **Lipids** are **transported** to adipose tissue in the form of **triglycerides,** and they are delivered to user tissues in the same form.
- At the level of **the capillaries**, chylomicrons and VLDLs are stripped of their triglycerides by a **lipoprotein lipase**, which hydrolyses them into **fatty acids** and **2-monoacylglycerol**.
- **The fatty acids** and **monoglyceride** enter adjacent cells: whether muscle or fat cells, these compounds are utilized:

-Directly in **β-oxidation**

-Or are **converted** back into **triglycerides** for **storage**.



B/Mobilization of stored triglycerides

-Stored triglycerides are a source of energy that can be utilized by all cells.

-They are **mobilized** in the **absence of glucose**.

-Prolonged fasting, physical exercise and stress promote their **mobilization**.

-They are hydrolyzed by a **triglyceride lipase sensitive to hormones** (adrenaline, glucagon, noradrenaline, corticosteroids, pituitary hormones, etc.).

-Their hydrolysis in adipocytes yields **fatty acids and 2-monoacylglycerol.**

Metabolism of fatty acids

I/Lipolysis: is a set of energy-producing metabolic pathways that enable the phosphorylation of ADP to ATP through the oxidation of fatty acids.

The **substrate** passes through metabolic junctions:

- Acyl-coenzyme A,
- Acetyl-coenzyme A (CH3~CO~SCoA),
- Hydrogen-carrying coenzymes.

The following **metabolic pathways** are distinguished among these junctions:

- **β-oxidation**, from **acyl-CoA** to **acetyl-CoA**.
- **Krebs cycle**, from **acetyl-CoA** to **bicarbonate**;
- **Mitochondrial respiratory chain**, from **hydrogen-carrying coenzymes** to **water** and from **ADP to ATP.**

β-oxidation:

A/ Definition:

-This is a metabolic pathway enabling the **oxidation of acyl-coenzyme A** in the **cytoplasm** to acetyl-coenzyme A in the presence of coenzymes that transport hydrogen to the respiratory chain.

-Oxidation of fatty acids = β-oxidation = Lynen cycle

-Occurs in many tissues such as muscles, the liver, the kidneys, adipocytes, etc. **except those that are glucose-dependent**, such as **the brain** (fatty acids do not cross the blood-brain barrier) and **erythrocytes** (which lack mitochondria).

-β-oxidation of fatty acids takes place in the **mitochondrial matrix**, where the β-oxidation enzymes are located near the respiratory chain complexes.

-β-oxidation occurs at the **β-carbon** (on the aliphatic chain corresponding to carbon **number 3** of the fatty acid) and involves the successive hydrolysis of **two carbon atoms.**

Acide gras saturé à n carbones

$$ {}_3HC_{\omega}\text{-}(CH_2)_n\text{-}{}^3C_{\beta}H_2\text{-}{}^2C_{\alpha}H_2\text{-}{}^1COOH $$

B/Activation of fatty acids

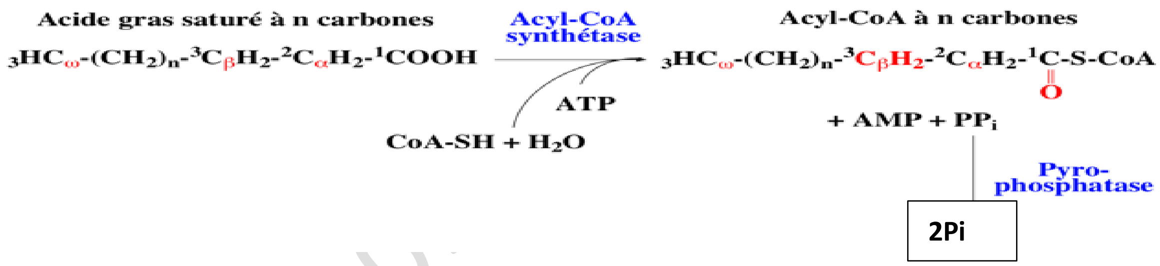
-The **first stage** of β-oxidation is **the activation** of fatty acids in **the cytoplasm**

-Fatty acids are **converted** into **acyl-CoA**.

-Activation requires **the hydrolysis of ATP**, consuming **two high-energy bonds**

-This reaction is catalyzed by **acyl-CoA synthetase** or **acyl-thiokinase**.

R-CH2-COOH + ATP + HSCoA ⟶ R-CH2-CO~SCoA + AMP + PPi

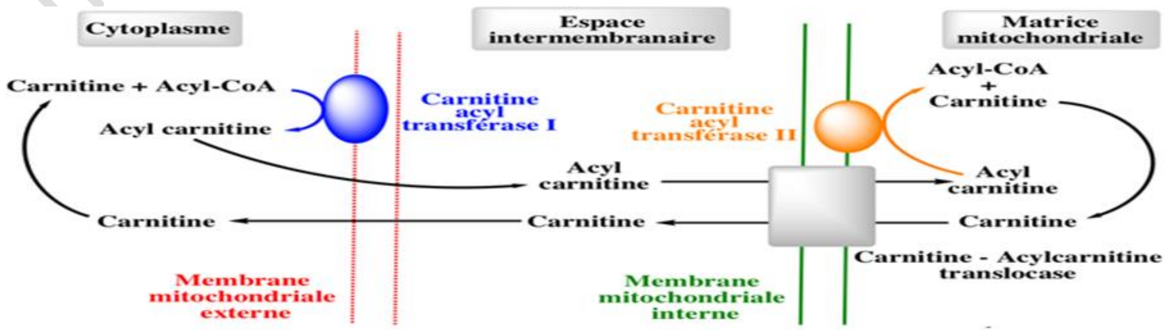


C/Entry into the mitochondrion: the role of carnitine

-Mitochondrial membranes are **impermeable** to **long-chain acyl-CoA**.

-They cross the membrane by being converted into **acyl-carnitine** from carnitine.

Acyl-CoA + Carnitine ⟶ Acyl-carnitine + HSCoA



Entry into the matrix occurs via:

- a shuttle system: **carnitine acyltransferases I and II**, located on the outer and inner membranes of the mitochondrion.

- a transport system: **the carnitine-acylcarnitine translocase**, located on the inner membrane of the mitochondrion.

D/Stages of β-oxidation of even-chain saturated fatty acids:

*The sequence of reactions takes place in **four stages**, known as **cycles**.

*For a fatty acid with **2n carbon atoms**, **(n-1) cycles** are required for its complete oxidation to **n acetyl-CoA.**

1/First dehydrogenation of acyl-CoA

-A **dehydrogenation** reaction occurs between **carbons 2 and 3** of acyl-CoA.

-The enzyme involved is **acyl-CoA dehydrogenase**, a **FAD-containing flavoprotein**, which creates a **double bond.**

-The product is **trans-Δ^2-enoyl-CoA**.

$$R\text{-}CH_2\text{-}CH_2\text{-}CH_2\text{-}CO\sim SCoA + FAD^+ \longrightarrow R\text{-}CH_2\text{-}CH{=}CH\text{-}CO\sim SCoA + FADH_2$$

2/Hydration of the double bond

-This is carried out by an **enoyl-CoA** (dehydroacyl-CoA) **hydratase**.

-The product formed is **3-hydroxyacyl-CoA**.

-The **OH** group is attached to carbon **3 or β**.

$$R\text{-}CH2\text{-}CH{=}CH\text{-}CO\sim SCoA + H2O \longrightarrow R\text{-}CH2\text{-}CHOH\text{-}CH2\text{-}CO\text{-}SCoA$$

3/Second dehydrogenation

-This involves **3-hydroxyacyl-CoA**.

-The hydrogen acceptor is **NAD+.**

-Oxidation of the alcohol group leads to a ketone group.

-The enzyme is **3-hydroxyacyl-CoA dehydrogenase.**

-The resulting compound is **3-ketoacyl-CoA:**

$$R\text{-}CH2\text{-}CHOH\text{-}CH2\text{-}CO\sim SCoA + NAD^+ \longrightarrow R\text{-}CH2\text{-}CO\text{-}CH2\text{-}CO\sim SCoA + NADH,H^+$$

4/Fatty acid cleavage

-This is the **final reaction** in the sequence.

-The enzyme involved is **ß-ketothiolase (lyase).**

-During thiolysis in the presence **of HSCoA, acetyl-CoA** is released and an **acyl-CoA** is reformed, with **two carbon atoms missing** from its chain.

-This acyl-CoA will serve as a **substrate** for the **next cycle**.

R-CH2-CO-CH2-CO~SCoA + HSCoA ⟶ R-CH2-CO~SCoA + CH3~CO~SCoA

- When only 4 carbon units remain**, β-ketothiolase** releases 2 acetyl-CoA.
- At the end of each cycle, 1 acetyl-CoA, 1 FADH2 and 1NADH,H+ are released into the matrix.
- If we start with a fatty acid containing **2n** carbon atoms, **(n-1)** cycles are required to achieve complete β-oxidation of the fatty acid, with the release **of n acetyl-CoA.**

E/The energy balance:

The overall reaction for the β-oxidation of a **fatty acid** with **2n carbon** atoms is written as:

(Acyl CoA) **2n** + **(n-1)** CoA + **(n-1)**FAD + **(n-1)**NAD+ + **(n-1)** H_2O ⟶ **n** Acetyl CoA + **(n-1)**$FADH_2$ + **(n-1)** NADH + **(n-1)**H^+ .

Example 1: C18:0

Stearate (C 18=2n=2×9) is activated to form stearyl CoA in exchange for 2 ATP.

The latter, after 8 cycles (n-1=9-1=8) of the cycle, yields 9 Acetyl CoA.

Stearyl CoA + 8 CoA + 8 FAD + 8 NAD+ + 8 H2O → 9 Acetyl CoA + 8 FADH2 + 8 NADH + 8 H+.

9 acetyl-CoA → 9 × 12 ATP = 108 ATP (in the Krebs cycle)

8 $FADH_2$ → 8 × 2 ATP = 16 ATP

8 NADH → 8 × 3 ATP = 24 ATP   40 ATP

2 phosphate bonds were consumed for the activation of the **AG (-2 ATP)**

108 + 40 = 148

The complete oxidation of one stearyl-CoA therefore yields **148 ATP**.

The complete oxidation of one stearate yields **146 ATP**, as 2 ATP are used to convert stearyl-CoA to stearate.

Net yield: **146 ATP**

Example 2: C16:0

Palmityl CoA + 7 CoA + 7 FAD + 7 NAD+ + 7 H_2O → 8 acetyl CoA + 7 $FADH_2$ + 7 NADH + 7 H^+.

8 acetyl CoA → 8 × 12 ATP = 96 ATP

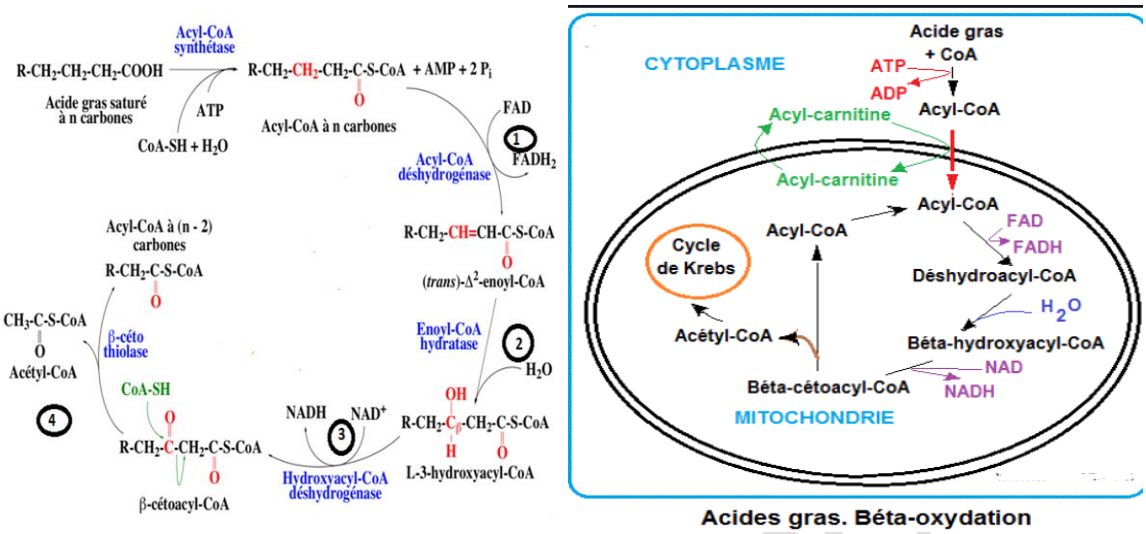
7 $FADH_2$ → 7 × 2 = 14 ATP

7NADH → 7 × 3 = 21 ATP   35 ATP

35 + 96 = 131 ATP.

The **complete oxidation of one palmityl CoA** therefore yields **131 ATP**.

The **complete oxidation of one palmitate** yields **129 ATP**, as 2 ATP are used in the conversion from palmityl CoA to palmitate.



The pathway of beta-oxidation of saturated fatty acids

F/Special case:

❖ **β-oxidation of odd-chain saturated fatty acids:**

Their oxidation follows the same steps as that of even-chain fatty acids.

Indeed, **the final step** in the β-oxidation of a fatty acid with an **odd number** of carbon atoms yields **acetyl-CoA (2 carbon atoms) and propionyl-CoA (3 carbon atoms).**

Propionyl-CoA is converted into **succinyl-CoA**, an intermediate in the Krebs cycle.

❖ **The β-oxidation of unsaturated fatty acids:**

-Monounsaturated fatty acids are broken down in the same way as saturated fatty acids after their activation and binding to coenzyme A.

-However, **two enzymes,** an **isomerase** and an **epimerase**, are required for the complete oxidation of these acids.

An isomerase:

-Oleic acid (C18) has a cis double bond between carbons 9 and 10.

-The first three rounds remove 6 carbons in the form of 3 acetyl-CoA molecules.

-The remaining molecule has a double bond between C3 and C4 in the cis configuration, which prevents the formation of the β-oxidation double bond between C2 and C3.

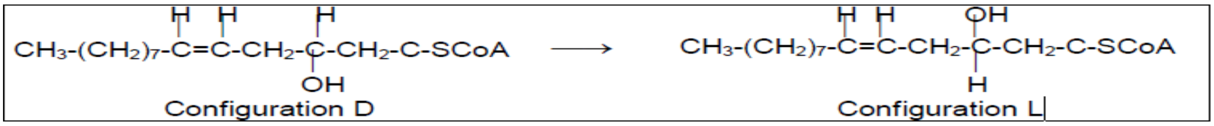
-The **isomerase** converts the cis bond to trans and shifts it between **C2 and C3**, allowing β-oxidation to continue.

CH3-(CH2)7-CH=CH-CH2-CO-SCoA ⟶ CH3-(CH2)7-CH2-CH=CH-CO-SCoA

An epimerase:

-In the case of unsaturated compounds with cis double bonds at positions 6 and 9, the first two steps remove 2 acetyl-CoA.

- The remaining compound, which has two double bonds at positions 2 and 5, is hydrated at the first double bond, yielding a product of **the D-configuration** which is not a substrate for β-oxidation. It is then epimerized to **the L-compound** by an epimerase.



Regulation of β-oxidation

- **Metabolic regulation**

-The rate of fatty acid oxidation is determined by the rate at which acyl-CoA enters the mitochondria via **the activity of acyl-carnitine transferase** (the main site of β-oxidation regulation)

This rate can be modified by **malonyl-CoA** (the product of acetyl-CoA carboxylation), which **inhibits acyl-carnitine transferase 1**.

- **High ATP/ADP inhibits** the entry of NADH+ and FADH+ into the respiratory chain ⟶ ↑ in their concentration ⟶ inhibition of the action of the two β-oxidation dehydrogenases.

- **High acetyl-CoA** concentration ⟶ **inhibition** of **thiolase.**

- **Hormonal regulation: Insulin:**

It facilitates the transport of glucose into adipose tissue, which **stimulates lipogenesis** and **reduces lipolysis**; therefore, **insulin has an anti-lipolytic effect**.

Metabolic pathways of acetyl-CoA, glycerol and propionyl-CoA:

1 - Metabolic pathways of glycerol

Glycérol

↓ (Glycérol kinase + ATP)

Glycérol 3-P

├──→ ** Lipid synthesis **

↓ (Glycérol-P déshydrogénase + NAD^+)

3-P-dihydroxyacétone

↓ (Phosphotriose isomérase)

Glycéraldéhyde 3-P

├──→ **Glycolysis** (energy)

└──→ **Neoglucogenesis** (→ glucose)

2 - Fate of propionyl-CoA:

Propionyl-CoA

↓ (Propionyl-CoA carboxylase + ATP)

2-Methylmalonyl-CoA

↓ (Méthylmalonyl-CoA mutase)

Succinyl-CoA

├──→ ** **Krebs Cycle** ** (ATP)

└──→ ****Neoglucogenesis**** (via malate)

3-Fate of acetyl-CoA:

Acetyl-CoA

├──→ **Krebs cycle** → CO_2 + H_2O + ATP

└──→ **Synthesis**:

├─ Ketone bodies (alternative energy source)

├─ Cholesterol (hormones, cell membranes)

├─ Isoprenoids (various metabolites)

└─ Fatty acids (TG)

II/Lipogenesis: this refers to the set of metabolic pathways that enable the synthesis of triglycerides (TG) from acetyl-CoA. This process takes place in the **cytoplasm** (not in the mitochondria).

A/Introduction

❖ The biosynthesis of fatty acids and lipids serves **two purposes** within the cell:

1– Providing the fatty acids necessary for the **synthesis of structural lipids**;

2– Storing energy; when food is too rich and exceeds the body's needs, lipids are **stored in adipose tissue**.

❖ Fatty acid synthesis takes place entirely in the **cytosol**

❖ Lipid synthesis, like all biosynthesis, requires:

– Energy supplied by **ATP**,

– Reducing power, supplied in the form of **NADPH**, with H+ derived mainly from the **pentose phosphate pathway**,

– The **only precursor** for fatty acid synthesis is **acetyl-CoA**.

❖ Acetyl-CoA is derived from:

– the ß-oxidation of **fatty acids** (intramitochondrial),

– the oxidation of **pyruvate** (mitochondrial),

– the oxidative breakdown of so-called **ketogenic amino acids**.

B / Transfer of the acetyl group from the mitochondria to the cytosol.

This occurs in two phases:

1) Mitochondrial phase:

- Pyruvate imported from the cytosol is carboxylated by pyruvate carboxylase to form oxaloacetate.

  Pyruvate + CO_2 + ATP ⟶ **oxaloacetate** + ADP + Pi

- Oxaloacetate condenses with acetyl-CoA to form citrate (the first reaction of the Krebs cycle, catalyzed by citrate synthase).

  Oxaloacetate + **acetyl-CoA** + H_2O ⟶ **citrate** + HSCoA

- Citrate is transported across the inner mitochondrial membrane via **citrate translocase**.

2) Cytosolic phase:

-Under the action of an ATP-dependent **citrate synthase** and in the presence of **HSCoA**, citrate is cleaved into **acetyl-CoA** and **oxaloacetate**, which regenerates **pyruvate**.

The sequence of reactions is as follows:

- **citrate** + HSCoA + ATP ⟶ **oxaloacetate** + **acetyl-CoA** + ADP + Pi

  (citrate synthase)

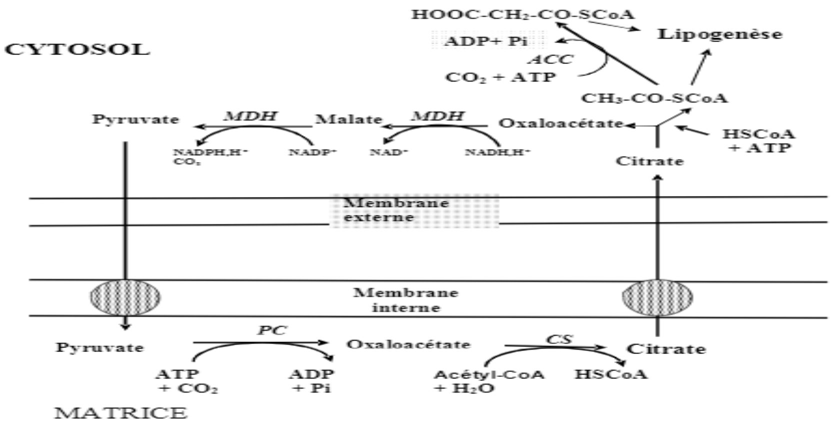
- **Oxaloacetate** + NADH, H+ ⟶ **malate** + NAD+

  (malate dehydrogenase to NAD+)
- **Malate** + NADP+ ⟶ **Pyruvate** + CO2 + NADPH, H+

  (malate dehydrogenase to NADP+)

-The regeneration of **pyruvate** allows the formation **of NADPH, H+,** which can also be used as a **reducing** agent in reactions catalyzed **by reductases**.

-The transport of **the acetyl group** from the matrix to the cytosol consumes **two high-energy phosphate bonds.**



C/ Biosynthesis of palmitic acid

-The synthesis of fatty acids is halted in the cytosol at the stage of **palmitic acid.**

-It requires the formation of **malonyl-CoA**, which donates **the two carbon atoms** during chain elongation, and the binding of intermediate acyl radicals to a carrier, known as **HSACP** (Acyl Carrier Protein),

1 – Molecules involved in palmitate synthesis

1.1 – **ACP-SH** (Acyl Carrier Protein)

1.2 - Formation of **malonyl-CoA**

CH3-CO~SCoA + **CO2** + ATP ⟶ **HOOC-CH2-CO~SCoA** + ADP + Pi

Acetyl-CoA Malonyl-CoA

1.3 - Transfer of **acetyl** and **malonyl** groups to **HSACP**

-The intermediate metabolites involved in palmitate synthesis are attached to HSACP in the cytosol by an acyltransferase: **acetyl transferase** and **malonyl transferase**.

-The reactions are as follows:

CH3-CO~SCoA + HSACP ⟶ **CH3-CO~SACP** + HSCoA

Acetyl-CoA (acetyltransferase) Acetyl-ACP

HOOC-CH2-CO~SCoA + HSACP ⟶ **HOOC-CH2-CO~SACP** + HSCoA

Malonyl-CoA (malonyl transferase) Malonyl-ACP

2 - Enzymatic steps in palmitate synthesis

2.1 - Condensation of acetyl-ACP and malonyl-ACP

-This reaction is catalyzed by **acetoacetyl-ACP synthase** (acyl-malonyl condensing enzyme), which is an enzyme that condenses these two molecules.

-The acceptor substrate is **acetyl-ACP**.

-Whereas the radical donor substrate is **malonyl-ACP**. The latter acts as the donor whenever chain elongation occurs.

-The catalyzed reaction is:

CH_3-CO~SACP + **HOOC-CH_2-CO~SACP** $\longrightarrow$ **CH_3-CO-CH_2-CO~SACP** + CO_2 + HSACP

Acetyl-ACP Malonyl-ACP Acetoacetyl-ACP

2.2 - Reduction of acetoacetyl-ACP to ß-hydroxybutyryl-ACP

-This reduction occurs in the presence of **NADPH** and H+ as electron and proton donors.

-It is catalyzed by **acetoacetyl-ACP reductase** (ß-ketoacyl-ACP reductase)

-The reaction is:

CH_3-CO-CH_2-CO~SACP + NADPH, H+ $\longrightarrow$ CH_3-CHOH-CH_2-CO~SACP + NADP+

Acetoacetyl-ACP **ß-hydroxybutyryl-ACP**

2.3 - Dehydration of ß-hydroxyacyl-ACP

-The enzyme responsible is ß-hydroxyacyl-ACP dehydratase, resulting in the formation of a double bond.

-This yields 2-enoyl-ACP

CH_3-CHOH-CH_2-CO~SACP $\longrightarrow$ CH_3-**CH=CH**-CO~SACP + H_2O

ß-hydroxybutyryl-ACP **2-enoylACP**

2.4 - Reduction of the double bond by NADPH, H+

-Unlike everything we have seen so far, the reduction of the double bond occurs in the presence of **NADPH and H+,** which supply the necessary electrons.

The enzyme is an **enoyl reductase**.

CH_3-CH=CH-CO~SACP + NADPH, H+ $\longrightarrow$ **CH_3-CH_2-CH_2-CO~SACP** + NADP+

2-enoyl-ACP **acyl-ACP 4C**

- The sequence of the last four reactions we have just seen:

Condensation, Reduction, Dehydration and Reduction $\rightarrow$ constitutes **one cycle**.

- The fatty acid we have just synthesised has **four carbon atoms**.
- It will become the radical-accepting substrate and its aliphatic chain will be extended by **2 carbon atoms** supplied by **malonyl-ACP** during the second cycle.
- This process will continue until **palmityl-ACP** is reached, which marks the end of fatty acid synthesis in the **cytosol**.
- For **longer-chain fatty acids**, elongation continues in the **mitochondria** and microsomes, which involves the transport of the **palmityl radical** into the mitochondria across the inner membrane via the **acyl-carnitine shuttle**, however, in the mitochondria, the donor of the acetyl group is then **acetyl-CoA.**

2.5 - The energy balance:

-The synthesis of **palmitic acid** is completed after **7 cycles** (which corresponds to **(n-1) cycles** for a fatty acid with **2n carbon atoms**).

-The overall reaction is as follows:

Acetyl-ACP + 7 malonyl-ACP + 14 (NADPH, H+) $\longrightarrow$ Palmitate + 8 HSACP + 14 NADP+ +7CO2

-Let us now establish this balance with **acetyl-CoA** as the **sole precursor**.

-We obtain the following sequence:

1/Acetyl-ACP + 7 malonyl-ACP + 14 (NADPH, H+) $\longrightarrow$ Palmitate + 8 HSACP + 14 NADP+ + 7 CO2

2/Acetyl-CoA + HSACP $\longrightarrow$ Acetyl-ACP + HSCoA

3/7 malonyl-CoA + 7 HSACP $\longrightarrow$ 7 malonyl-ACP + 7 HSCoA

4/7 Acetyl-CoA + 7 CO2 + 7 ATP $\longrightarrow$ 7 malonyl-CoA + 7 ADP + 7 Pi

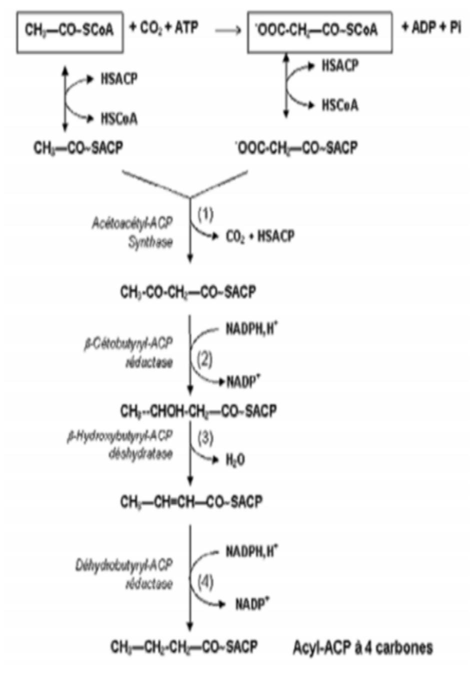
When the **four reactions** above are added together, we obtain:

8 Acetyl-CoA + 7 ATP + 14 (NADPH, H+) $\longrightarrow$ **Palmitate** + 8 HSCoA + 7 ADP + 7 Pi + 14 NADP+

2.6-Regulation of FA synthesis

-FA synthesis occurs when carbohydrates and energy are abundant, and when FAs are scarce.

-**Acetyl-CoA carboxylase** is the **key enzyme controlling** the pathway



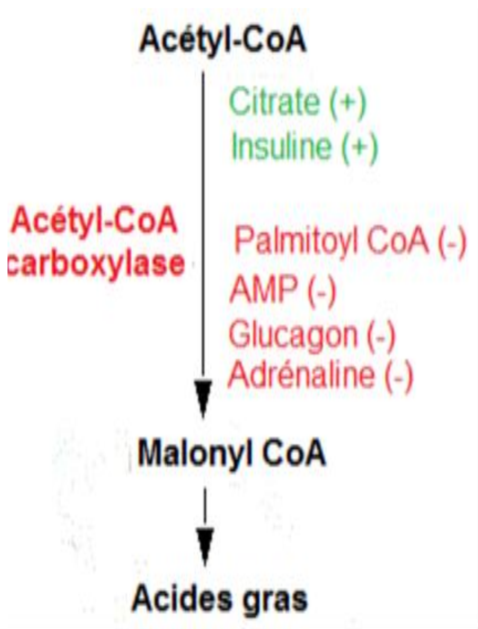


Diagram of the biosynthesis of palmitic acid **Regulation of FA synthesis**